TECHNICAL ROADMAP: JOINT UAV TRAJECTORY AND RIS OPTIMIZATION FOR AOI-AWARE DATA COLLECTION

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Subject: Implementation plan and technical roadmap for the Phase 2 research extension

1. Research Context and Novelty

Building upon our accepted WCNC 2026 methodology, the goal of this extension is to transition from a static downlink sum-rate maximization paradigm to a dynamic, IoT-centric uplink data collection framework. The core objective shifts to minimizing the Age of Information (AoI) while actively incorporating Unmanned Aerial Vehicle (UAV) 3D mobility constraints.

A comprehensive review of the 2024–2026 literature reveals a distinct research gap. While substantial work exists on UAV trajectory optimization for AoI minimization (often utilizing DDPG or PPO algorithms), and the integration of Reconfigurable Intelligent Surface (RIS) technology to aid UAV communications is well-established, the specific intersection of these domains under realistic hardware constraints remains entirely unexplored. The vast majority of current studies assume ideal RIS elements characterized by unit-amplitude reflection coefficients regardless of the applied phase shift.

Our previous work demonstrated that this ideal assumption leads to significant performance degradation due to Phase-Dependent Amplitude (PDA) mismatch. Therefore, optimizing UAV trajectory and scheduling for AoI minimization while explicitly accounting for the non-linear PDA characteristics of the RIS constitutes a highly novel, four-way research intersection (UAV + RIS + AoI + PDA) with strong potential for top-tier IEEE journal publication.

2. Extended System Model

We preserve the foundational DBSCAN-based clustering architecture established in our preliminary work. We consider a 100×100m urban area containing 80 IoT devices segregated into 10 clusters.

2.1 UAV Mobility and Energy Consumption

The UAV operates as a mobile aerial relay equipped with an L-element RIS. Its continuous 3D trajectory over a mission duration T is discretized into time slots Δt. The position at slot t is denoted by $\mathbf{q}(t) = [x(t), y(t), z(t)]^T$. The UAV is subject to practical aerodynamic constraints:

$$||\mathbf{q}(t+1) - \mathbf{q}(t)|| \leq V_{\max} \Delta t$$

We adopt a rigorous rotary-wing UAV propulsion energy model derived by Zeng and Zhang (IEEE TWC, 2019). The instantaneous power consumption $P(v)$ as a function of the UAV speed $v(t)$ encompasses blade profile power, induced power, and parasite power. The total energy expended must satisfy the battery capacity E_{budget}.

2.2 Age of Information Evolution

Data transmission occurs in the uplink from the ground devices to the UAV-mounted RIS, and subsequently to a central base station. Let $s_k(t) \in \{0, 1\}$ represent the scheduling indicator for device k. The continuous AoI evolution $\Delta_k(t)$ is defined as:

$$\Delta_k(t+1) = \begin{cases} 1, & s_k(t) = 1 \text{ and } R_k(t) \geq R_{\min} \\ \Delta_k(t) + 1, & \text{otherwise} \end{cases}$$

where $R_k(t)$ is the achievable uplink rate given the current channel state and RIS configuration.

2.3 Realistic Channel and PDA Model

To accurately characterize the air-to-ground communication links subject to UAV mobility, we replace the previous Rayleigh fading assumption with a distance-dependent Rician fading model. This incorporates a strong Line-of-Sight (LoS) component parameterised by the Rician K-factor K_{rice}.

Crucially, we retain the exact PDA model from the original work. The amplitude coefficient β_l is intricately coupled to the phase shift θ_l via the non-linear relationship:

$$\beta_l(\theta_l) = (1 - \beta_{\min}) \left(\frac{\sin(\theta_l - \bar{\theta}) + 1}{2} \right)^{\bar{\kappa}} + \beta_{\min}$$

3. Problem Formulation and Architecture

The global optimization objective is the minimization of the time-average AoI across all devices over the mission duration:

$$\min_{\{\mathbf{q}, \mathbf{\Phi}, \mathbf{G}, \mathbf{s}\}} \quad \frac{1}{K} \sum_{k=1}^{K} \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^{T} \Delta_k(t) \right]$$

This formulation yields a highly non-convex, mixed-integer non-linear programming (MINLP) problem. To tackle the curse of dimensionality and ensure stable convergence, we propose a Hierarchical Deep Reinforcement Learning (DRL) architecture.

3.1 High-Level Macro-Planner (PPO)

The macro-planner operates at a lower temporal resolution to manage long-term spatial dependencies. Utilizing Proximal Policy Optimization (PPO), it observes the global state $\mathcal{S}_{\text{high}} = \{\mathbf{q}(t), \mathbf{\Delta}(t), E_{\text{rem}}(t)\}$. The network outputs a hybrid action tuple determining the next target cluster ID (discrete) and the necessary continuous velocity vector $\mathbf{v}(t)$. The reward function is strictly the negative average AoI.

3.2 Low-Level Micro-Optimizer (Beta-Space SAC)

Operating at high resolution within the vicinity of a targeted cluster, the micro-optimizer utilizes the Beta-Space Exploration Soft Actor-Critic (SAC) algorithm previously developed. It ingests the instantaneous channel state information $\mathcal{S}_{\text{low}} = \{\mathbf{H}_1(t), \mathbf{H}_2(t)\}$ and continuously optimizes the RIS phase matrix $\mathbf{\Phi}(t)$ and BS receive beamforming $\mathbf{G}(t)$ to maximize localized sum-rate, thereby satisfying the minimum rate requirement $R_{\min}$ for the scheduled devices to successfully reset their AoI.

4. Implementation Strategy and Timeline

The execution of this roadmap will span five distinct phases over an 8-week horizon. Phase 1 (Environment Extension) has been completed, encompassing the integration of the Zeng & Zhang energy model, Rician channels, and AoI tracking logic within the `AoIUAVRISEnvironment` class.

Phase	Task Description	Estimated Duration
Phase 1	Environment Extension (UAV Mobility, Energy Model, AoI Tracking)	Completed (Weeks 1-2)
Phase 2	High-Level PPO Agent Development (Hybrid Action Space)	Weeks 2-3
Phase 3	Hierarchical Integration (Curriculum Learning Strategy)	Weeks 3-4
Phase 4	Baseline Implementation (Greedy, Round-Robin, Fixed-UAV)	Weeks 4-5
Phase 5	Evaluation, Data Generation, and Manuscript Preparation	Weeks 5-8

Phase 2 involves the construction of the hybrid PPO agent, followed by Phase 3 which integrates both networks using a curriculum learning approach. Pre-trained weights from the WCNC 2026 experiments will initialize the low-level SAC agent to significantly accelerate convergence.

5. Target Venues and Discussion Points

We propose a phased publication strategy to balance rapid dissemination and high academic impact.

5.1 Questions for Discussion

Before proceeding deeply into Phase 2 development, guidance is requested on the following aspects:

1. Shall we go with the Hierarchical DRL approach, or would you prefer investigating a flat architecture first despite the dimensionality challenges?
2. Is there a specific baseline method (e.g., purely mathematical optimization via SDR) you feel is mandatory to include for this problem domain?